Sector 5 HW Video: Speaker Notes

ECS 1100 · Advanced Manufacturing, Biomanufacturing & Supply Chain · slides 1–24 (the appendix major guides have no notes)

Slide 1: Cover

Welcome to Week 12 and Sector 5: advanced manufacturing, biomanufacturing and supply chain.

Almost everything you own passed through this sector. This week we follow a product from idea to your hands, and see where every Jonsson School major fits along the way. Let's start with our goals.

Slide 2: Learning objectives

By the end of the week, you should be able to describe how a product goes from design to delivery, explain what makes biomanufacturing different, identify where your major fits, and explain how one breakdown can affect the whole system.

Keep that last one in mind. To make this concrete, let's start with something familiar.

Slide 3: How everyday things get made

Take any object near you. Before it reached you, someone designed it, planned how to make it at scale, built it, checked it and shipped it.

A truck is welded by robots from supplier parts. A vaccine is grown in living cells and filled into sterile vials. A phone starts as chips in a fab.

That chain is this sector, and if any one link fails, the product never arrives. The chain has three branches.

Slide 4: One sector, three branches

Advanced manufacturing is making things with new technology like robots, sensors and data. Biomanufacturing uses living cells as the factory, and it is how most modern medicines are made. Supply chain connects everything, from raw material to the customer.

So how big is this sector?

Slide 5: By the numbers

The U.S. has about 12.6 million manufacturing jobs, and manufacturing is close to 10 percent of GDP. The country will need 3.8 million more workers by 2033, and many of those jobs could go unfilled.

Texas is a big part of this. It has about a million manufacturing jobs and has been the top exporting state for 24 years in a row.

Next, let's see how the production system works.

Slide 6: From design intent to delivery

Here is the system in six stages: design intent, process planning, production, quality, logistics and delivery. Under each stage are the majors who often do the core work there.

Notice that software, data and security are not in one box. They run through every stage.

Now let's see what changes when the factory is a living cell.

Slide 7: The biomanufacturing branch

In biomanufacturing, cells are grown in a bioreactor, harvested, purified, filled into sterile vials, and tested batch by batch before release.

This is harder to control than machining a part. Cells vary, and one contamination can ruin a batch worth millions. Because you cannot inspect quality in at the end, every step is validated and recorded under FDA Good Manufacturing Practice. No record, no release.

Either way, the product still has to move. That is the supply chain.

Slide 8: Supply chain: the system between the factories

The supply chain connects the factories. Materials flow forward from suppliers to customers, while data and forecasts flow back to tell factories what to make next.

Logistician jobs are projected to grow 18 percent, and North Texas, with DFW Airport and AllianceTexas, is a major logistics hub.

More and more of this work is automated, so let's see where.

Slide 9: Where automation, robots and vision enter

Automation enters at specific points: robotic cells in production, machine vision in quality, mobile robots in logistics, and digital twins, which are software models of the line, across all stages.

The U.S. installed about 38,000 industrial robots in 2025. China installed 295,000 the year before.

All of this takes many disciplines, which brings us to your major.

Slide 10: All ten Jonsson School programs have a role

All ten Jonsson School programs have a role here. Mechanical, Biomedical, Systems, Data Science and Software Engineering do core work, which we call primary. The other five have significant, named roles, from robotic controls to securing factory networks.

And many of these jobs are in our own state.

Slide 11: Why Texas is building factories

Texas is where many new U.S. factories are going. TI started production at its Sherman chip fab in December 2025, Samsung is building in Taylor, SpaceX makes Starlink hardware in Bastrop, and Apple is building AI servers in Houston.

In our job log, 71 of 100 postings are in Texas, and 21 are in DFW. Biomanufacturing is growing here too.

Slide 12: Texas biomanufacturing is growing too

Eli Lilly and Bristol Myers Squibb are investing billions in Houston, FUJIFILM is expanding in College Station, and Lonza has a cell and gene therapy site in Pearland.

For biomedical engineers, these companies hire process, validation and quality engineers, the same titles you will see in the job log.

So what is coming next?

Slide 13: What is exciting right now

A few things are moving fast. Apptronik in Austin has raised over 935 million dollars for factory humanoid robots. TI's Sherman fab is now running. Tesla says its Optimus line could build up to 10 million robots a year, though that is a company target. And digital twins are becoming standard.

You do not have to wait until graduation to get involved.

Slide 14: Manufacturing at UT Dallas

At UTD, you can work in the manufacturing lab, the NSERL cleanroom or the Biomedical Device Center, or take supply chain courses in the Jindal School. Student orgs like Comet Robotics and Formula Racing give you real build experience.

Now let's return to our big idea: what happens when one link breaks.

Slide 15: When one stage fails, the whole system feels it

In 2021, chip fabs were already at capacity, so automakers ran short. Lines stopped, half-built trucks sat parked, and U.S. vehicle inventory fell from 3.6 million to 1.5 million. A part worth a few dollars stopped entire plants.

The same is true of one contaminated bioreactor. On your concept map, ask of every node: what breaks if this one fails?

Sometimes, though, the failure is a human decision.

Slide 16: Two decisions on the factory floor

Here are two ethics cases. In the first, your automation project removes a shift of workers but also takes them out of a hazardous job. What do you owe them?

In the second, you are asked to sign a batch record before testing is done. In GMP manufacturing, accurate records are a legal duty, and the FDA has issued warning letters for falsified ones.

Bring your answer to the panel. Closely related is worker safety.

Slide 17: Worker safety is an engineering decision

Manufacturing still sees about 2.7 injuries per 100 workers. Engineers shape that number when they design a line: guarding hazards, automating with people in mind, and giving anyone the authority to stop an unsafe line.

Employers notice. Forty-four of our 100 postings ask for safety skills. Let me explain how that job log was built.

Slide 18: How the job log was built

I searched LinkedIn for recent postings, Texas first. I kept only jobs where the work is in manufacturing or supply chain and the posting accepts the major. I confirmed each was still open, kept ten per major, and saved each as a PDF.

That gave us 100 postings from 74 employers. Here is what they ask for.

Slide 19: What almost every posting asks for

The most common asks are not technical. Eighty percent want teamwork, 80 percent want communication, and two-thirds want problem solving. The top technical skill is automation and controls, at 58 percent.

The skills you practice in this course matter. Now let's break it down by major.

Slide 20: Top technical asks by major

Here are the top technical asks for each major. Biomedical is GMP and validation. Computer, CS and software engineering lean on automation and controls. Cybersecurity is OT security, and Data Science is data analysis and supply chain planning.

Find your major and note what you still need to build. All of these postings are in one place.

Slide 21: Where to find them all

The Sector 5 Job Explorer is linked on Canvas. You can filter by major, career stage or Texas, and each card links to the live posting and a saved PDF of the full text.

Once you find postings you like, UTD can help you get ready for them.

Slide 22: Career support on campus

Jonsson Career Services helps with resumes and internship searches, the Career Center runs career fairs, and UTDesign pairs you with company projects. LinkedIn Learning is free through the library.

This week, open three postings for your major, list the skills you are missing, and match each to a class, lab, org or free course.

You will also hear from people in the field at this week's panel.

Slide 23: Questions to bring to the panel

Come to the panel with at least one question. You could ask which stage they work in, what happened the last time a stage failed, how automation has changed their jobs, or what a new engineer does in month one. Or, tying back to the ethics case, ask if they have ever refused to sign something.

The panel will also help with your homework.

Slide 24: HW 12: Sector 5 concept map

For HW 12, build a concept map of both branches, from design intent to delivery. Label each node with the majors that act on it. Find the busiest node and the emptiest one, and decide if the empty one is real or a gap. Then connect the batch-record case to the quality node.

Submit it on Canvas by Sunday, November 15, at 11:59 PM. Thanks, and I look forward to seeing your maps.